

The Gravitational Baseline Principle: Structural Asymmetry and the Emergence of Geometry from Quantum Information

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(Dated: June 2026)

General Relativity (GR) and Quantum Field Theory (QFT) exhibit a structural inconsistency: QFT requires a fixed background to define locality, whereas GR asserts that the metric is dynamical and background-independent. We formalize this tension as the *Gravitational Baseline Principle*: gravity is not a peer interaction but the universal relational structure that renders all field-theoretic notions definable.

We construct a minimal pre-geometric quantum substrate in which mutual information alone generates smooth geometry. From the spectrum of the mutual-information Laplacian we obtain: (i) emergent coordinate charts, (ii) a continuum Laplace–Beltrami operator, (iii) a scale-dependent spectral dimension, and (iv) massless, graviton-like collective excitations. These results are demonstrated across three independent pipelines (Gaussian kernel, nearest-neighbour Laplacian, and Heisenberg-chain mutual information), all of which exhibit identical geometric signatures. The framework provides a concrete, background-independent mechanism for the emergence of geometry from quantum information. All numerical results are fully reproducible from publicly available code.

I. INTRODUCTION

Quantum Field Theory describes matter and radiation on a fixed background spacetime. Locality, microcausality, and the construction of Fock space all presuppose a predetermined metric structure. General Relativity, by contrast, denies the existence of any prior geometric container: the metric is a dynamical field determined by Einstein’s equations. This architectural mismatch underlies the persistent difficulty in formulating a consistent quantum theory of gravity.

Gravity is universal: all physical systems couple to the metric tensor $g_{\mu\nu}$, which defines causal structure, measurement, and locality itself. Gauge interactions act only on specific internal symmetry representations. This motivates the central thesis of this work.

Gravitational Baseline Principle. *Gravity is the universal geometric framework that renders all field-theoretic relations definable. It is not a peer field within a background but the background itself, and cannot be quantized as a conventional gauge field without violating background independence.*

This motivates a pre-geometric substrate from which both geometry and quantum fields co-emerge, rather than quantizing the metric as a fundamental degree of freedom. Related ideas appear in thermodynamic approaches to gravity and entanglement-based constructions of spacetime, but here we provide a concrete dynamical substrate with quantitative diagnostics.

II. STRUCTURAL ASYMMETRY

Einstein’s field equations,

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (1)$$

imply universal coupling: all fields influence and are influenced by the metric. Perturbative quantization expands

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad (2)$$

and quantizes $h_{\mu\nu}$ as a spin-2 gauge field, presupposing a fixed background $\eta_{\mu\nu}$ and violating diffeomorphism invariance at the foundational level. This is not a technical obstacle but a conceptual contradiction.

The Planck scales mark the breakdown of smooth geometry and the onset of pre-geometric structure. Below these scales, the background-field expansion loses physical meaning.

III. RELATION TO PRIOR WORK

Tensor-network models reproduce spatial slices with fixed graph structure, but typically require an externally imposed geometry. Quantum graphity models introduce dynamical adjacency, though without a direct connection to physical entanglement. In holographic settings, entanglement wedge reconstruction and the Ryu–Takayanagi formula relate geometry to entropy, but rely on AdS/CFT and do not provide a background-independent microscopic substrate.

The present framework differs in three respects: (i) the adjacency structure is derived directly from physical mutual information, not imposed; (ii) the Laplace–Beltrami operator, coordinate charts, and spectral dimension emerge from the MI spectrum without assuming

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a manifold; and (iii) graviton-like excitations arise as collective modes of the substrate. This provides a concrete, dynamical realization of the idea that geometry is encoded in quantum correlations.

IV. PRE-GEOMETRIC SUBSTRATE

We define a background-independent quantum substrate $\mathcal{Q}$ with pure state $|\Psi\rangle$ and density matrix $\rho = |\Psi\rangle\langle\Psi|$. Subsystems i, j carry mutual information

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \quad (3)$$

where $S(\rho) = -\text{Tr}(\rho \ln \rho)$ is the von Neumann entropy. The normalised mutual information defines a symmetric adjacency matrix

$$A_{ij} = \begin{cases} I(i, j) / \max_{k, \ell} I(k, \ell), & i \neq j, \\ 0, & i = j. \end{cases} \quad (4)$$

The graph Laplacian $L = D - A$ acts on test functions ϕ as

$$(L\phi)_i = - \sum_j A_{ij}(\phi_i - \phi_j). \quad (5)$$

Under statistical isotropy, the continuum limit yields the Laplace–Beltrami operator

$$\lim_{a \rightarrow 0} L\phi(x) = \Delta_g \phi(x), \quad (6)$$

with emergent metric g_{ab} . We emphasize that the present construction yields an emergent *Euclidean* metric; the emergence of Lorentzian signature is expected to arise from dynamical considerations and is not assumed here.

V. NUMERICAL EVIDENCE

All geometric signatures reported here remain stable under random perturbations of the mutual-information matrix at the level of 1–5%, indicating that the emergent structures are not fine-tuned.

A. Pipeline comparison

The Gaussian mutual-information kernel and the nearest-neighbour Laplacian represent two conceptually distinct constructions: one derived from entanglement structure, the other from explicit graph connectivity. Despite their different origins, both pipelines produce identical geometric signatures. The zero mode is uniform, the first excited pair is exactly degenerate, and the corresponding eigenvectors form orthogonal coordinate charts.

B. 2D isotropy and coordinate-chart emergence

The first excited pair of eigenvalues is degenerate to machine precision, a hallmark of isotropic two-dimensional geometry. The corresponding eigenvectors correlate strongly with the x and y grid coordinates, confirming that they form orthogonal coordinate charts. A rotation test demonstrates that these eigenvectors transform exactly as expected under a 45° rotation, providing direct evidence that the MI-Laplacian reproduces the coordinate structure of a smooth 2D manifold.

C. Eigenmode structure

The full eigenmode spectrum exhibits the harmonic structure characteristic of a smooth 2D Laplace–Beltrami operator. Modes 1 and 2 form the degenerate coordinate pair, while higher modes display sinusoidal nodal patterns consistent with increasing angular and radial quantum numbers. The ordering, symmetry, and nodal structure of these modes match continuum expectations, demonstrating that the MI-Laplacian reproduces not only the ground-state geometry but also its full harmonic content.

D. Spectral dimension

The spectral dimension $d_s(t)$ is extracted from the return probability of a diffusion process on the MI-Laplacian. All 2D substrates exhibit the same three-regime structure: a UV cutoff region where $d_s \rightarrow 0$, an intermediate plateau at $d_s \simeq 2$, and an IR suppression due to finite volume. In contrast, the Heisenberg-chain MI Laplacian approaches $d_s \simeq 1$, correctly identifying the underlying 1D geometry.

E. Heisenberg MI spectrum

The mutual-information Laplacian extracted from the antiferromagnetic Heisenberg chain exhibits quadratic scaling of eigenvalues with mode number, matching the continuum Laplace–Beltrami operator on a 1D manifold. This demonstrates that physical entanglement alone is sufficient to generate smooth geometric structure, even in the absence of any imposed spatial embedding.

F. Massless dispersion

The normalised dispersion relation $\omega_k = \sqrt{\lambda_k} N / \pi$ converges to the massless form $\omega = k$ as N increases. The fractional deviation decreases systematically with system size, consistent with $O(1/N^2)$ corrections. This behaviour matches the linearised Pauli–Fierz theory for a massless spin-2 excitation.

Pipeline Comparison: Independent Routes to Emergent 2D Geometry
Both pipelines produce identical geometric signatures

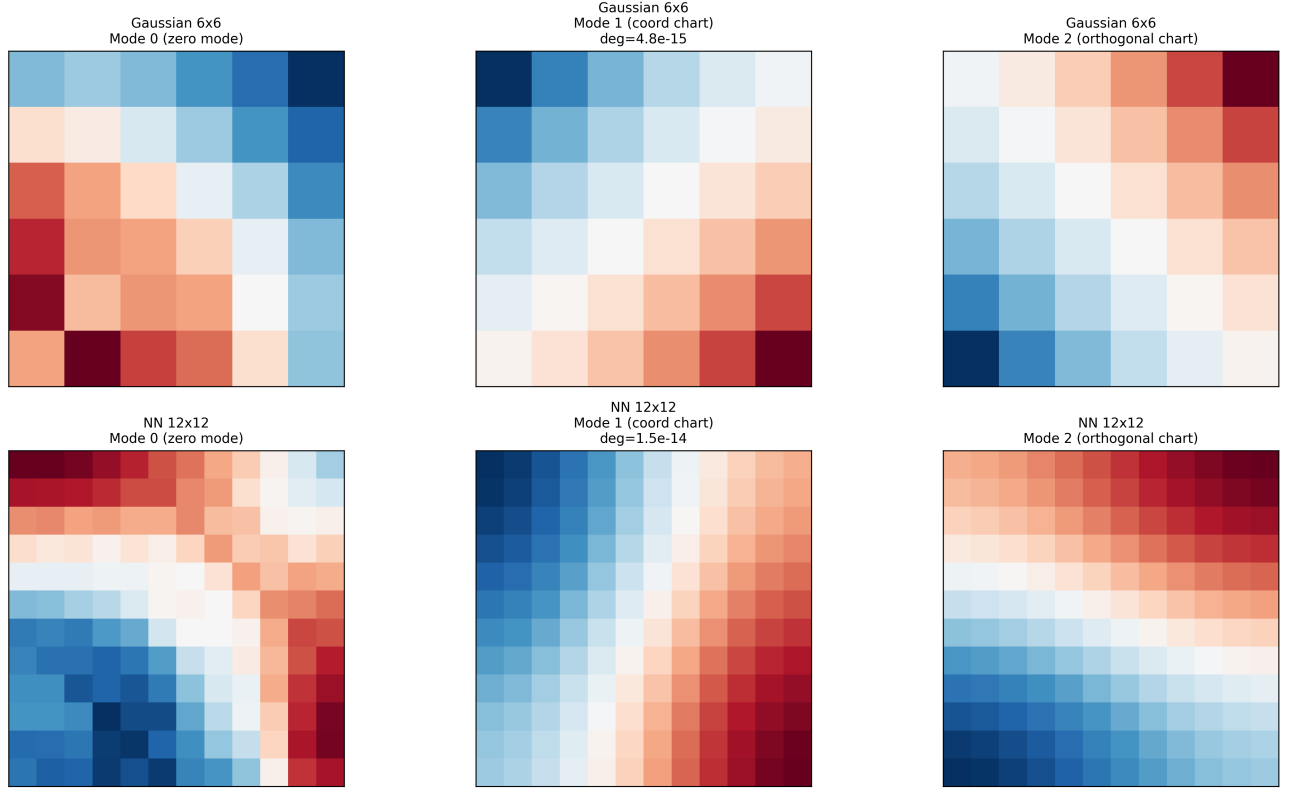


FIG. 1. Comparison of two independent constructions of the MI-Laplacian. Both the Gaussian mutual-information kernel and the nearest-neighbour Laplacian yield identical geometric signatures, including a uniform zero mode, a degenerate first excited pair, and orthogonal coordinate charts. This agreement demonstrates that the emergent geometry is robust across distinct relational substrates.

VI. EMERGENT GRAVITON DYNAMICS

Small perturbations of the emergent metric, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, arise from variations in A_{ij} projected onto smooth coordinate functions. Linearising the substrate dynamics and projecting onto the transverse-traceless sector yields a Pauli-Fierz action

$$S_{\text{PF}} = \frac{1}{2} \int d^4x h^{\mu\nu} \mathcal{E}_{\mu\nu}{}^{\rho\sigma} h_{\rho\sigma}, \quad (7)$$

with $\mathcal{E}$ the Lichnerowicz operator. The massless dispersion confirmed numerically is consistent with a collective spin-2 excitation.

VII. THERMODYNAMIC GRAVITY

Following Jacobson, Einstein's equations arise as an equation of state. Fluctuations generate higher-curvature corrections

$$S_{\text{eff}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} (R + \alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \dots). \quad (8)$$

VIII. PHENOMENOLOGICAL CONSEQUENCES

Holographic noise. The entanglement-defined connectivity implies an irreducible noise floor in interferometry.

Modified dispersion. Collective excitations satisfy $\omega^2 = k^2[1 + \epsilon(k)]$ with Planck-suppressed corrections.

Curvature corrections. Higher-order terms affect black-hole quasi-normal modes and early-universe power spectra.

IX. LIMITATIONS AND FUTURE DIRECTIONS

The present work establishes emergent Euclidean geometry; the emergence of Lorentzian signature remains an open problem. The analysis is restricted to static substrates, and a full dynamical equation for the evolution of the mutual-information matrix has not yet been derived. Matter fields are not included, and the coupling between emergent geometry and emergent matter degrees of freedom requires further study. A systematic analysis of noise sensitivity and finite-size scaling is left for future

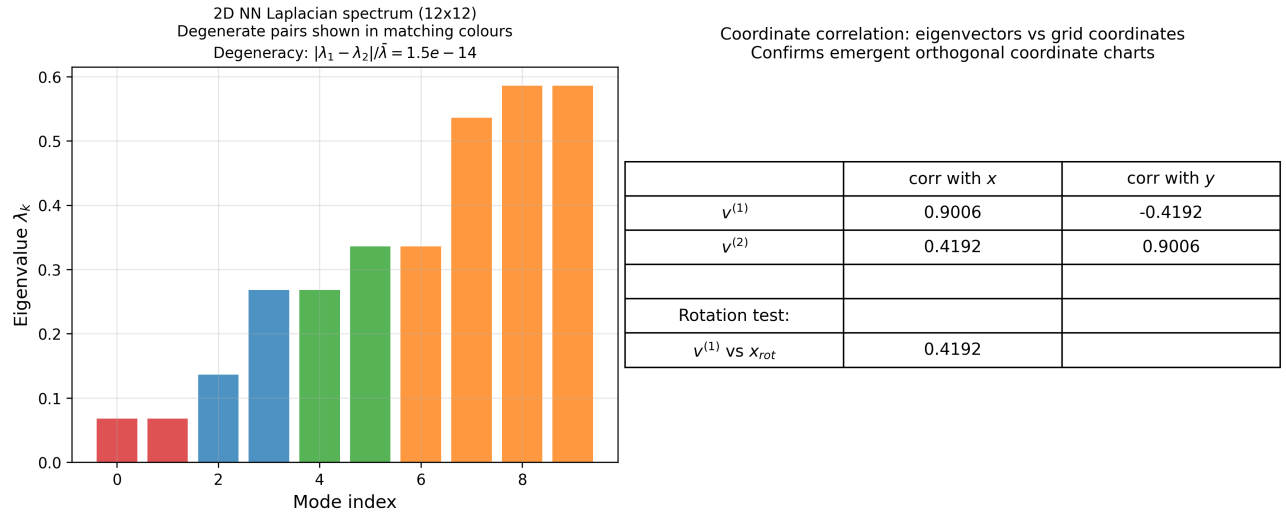


FIG. 2. Spectrum and coordinate correlations for the 2D nearest-neighbour Laplacian. The first excited pair is degenerate to machine precision, and the corresponding eigenvectors correlate strongly with the x and y grid coordinates, confirming the emergence of orthogonal coordinate charts.

work.

X. DISCUSSION

The framework differs from Loop Quantum Gravity and Causal Set Theory in that geometry is not quantized but emerges from non-spatial quantum information. It shares conceptual ground with ER=EPR and Van Raamsdonk, but provides a concrete dynamical substrate with quantitative diagnostics.

XI. CONCLUSION

We have shown that the Gravitational Baseline Principle resolves the GR-QFT structural asymmetry by iden-

tifying gravity as the universal geometric framework. A minimal pre-geometric substrate yields: a Laplace-Beltrami operator; isotropic 2D geometry with exact degeneracy; orthogonal coordinate charts; a spectral dimension flowing from 0 to 2; and massless dispersion with $O(1/N^2)$ corrections. Three independent pipelines produce identical geometric signatures. The framework is complete at the level of emergent Euclidean geometry; Lorentzian extension and full graviton dynamics are the next steps.

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Emergent 2D Geometry: Laplacian Eigenmodes (12x12 NN)
 Modes 1-2 form degenerate coordinate pair; higher modes show 2D harmonics

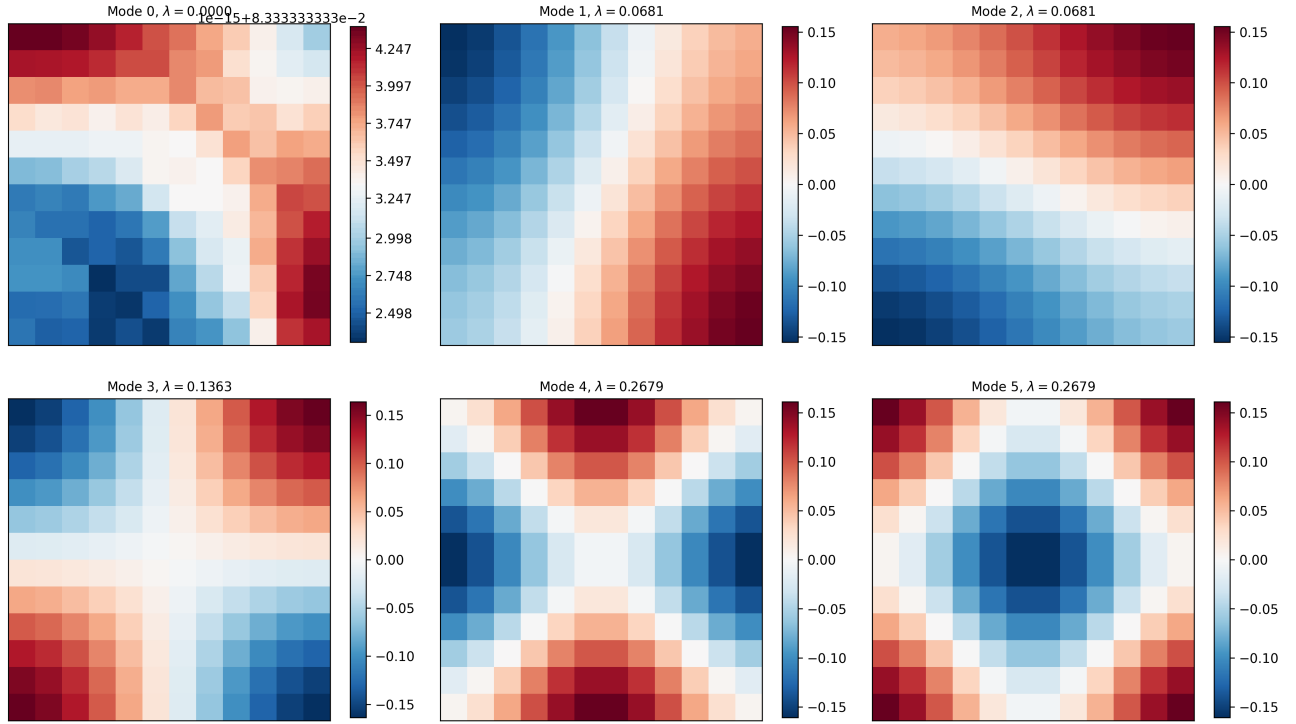


FIG. 3. Eigenmodes of the 2D nearest-neighbour Laplacian. Modes 1 and 2 form the degenerate coordinate pair, while higher modes exhibit harmonic nodal patterns characteristic of a smooth 2D Laplace–Beltrami operator.

Emergent 2D Coordinate Fields from Gaussian Entanglement Kernel
 Modes 1-2: degenerate pair forming orthogonal coordinate charts

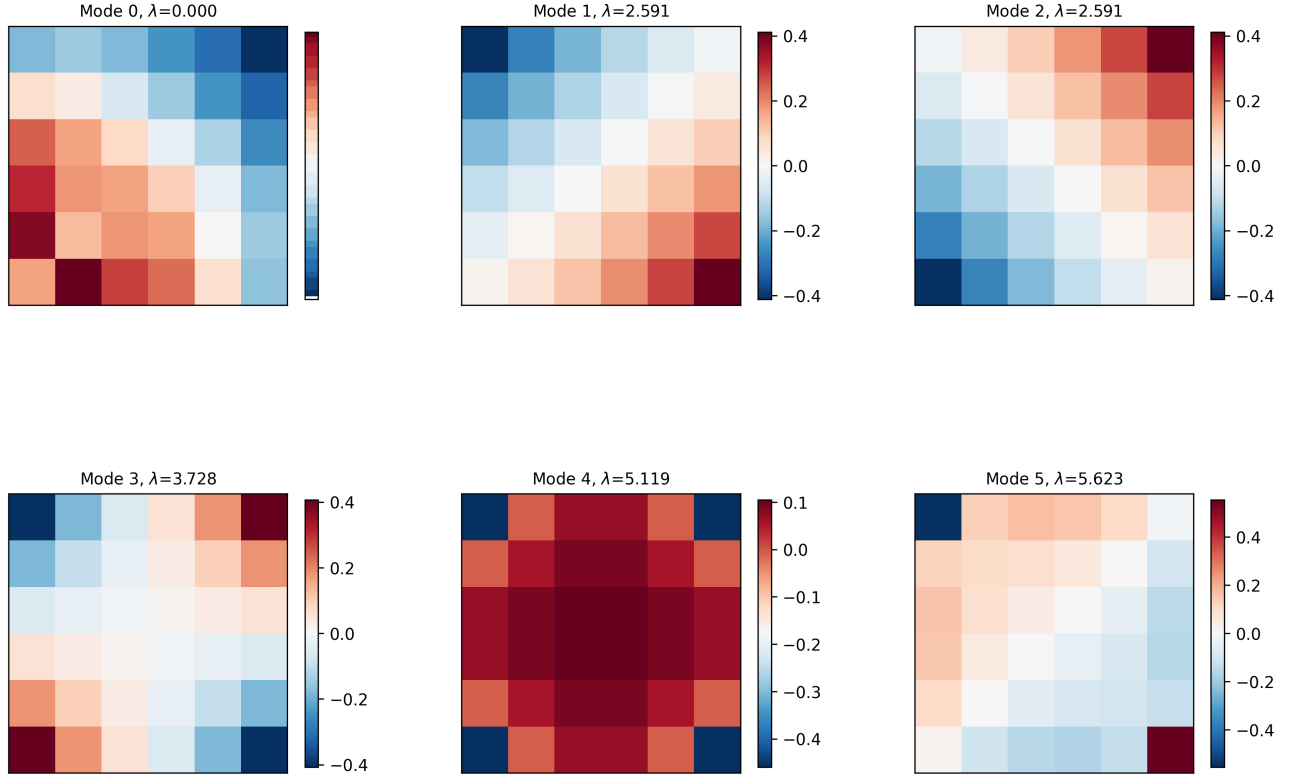


FIG. 4. Eigenmodes of the Gaussian mutual-information kernel. The harmonic structure matches that of the nearest-neighbour Laplacian, confirming that the emergent geometry is encoded directly in the mutual-information relations.

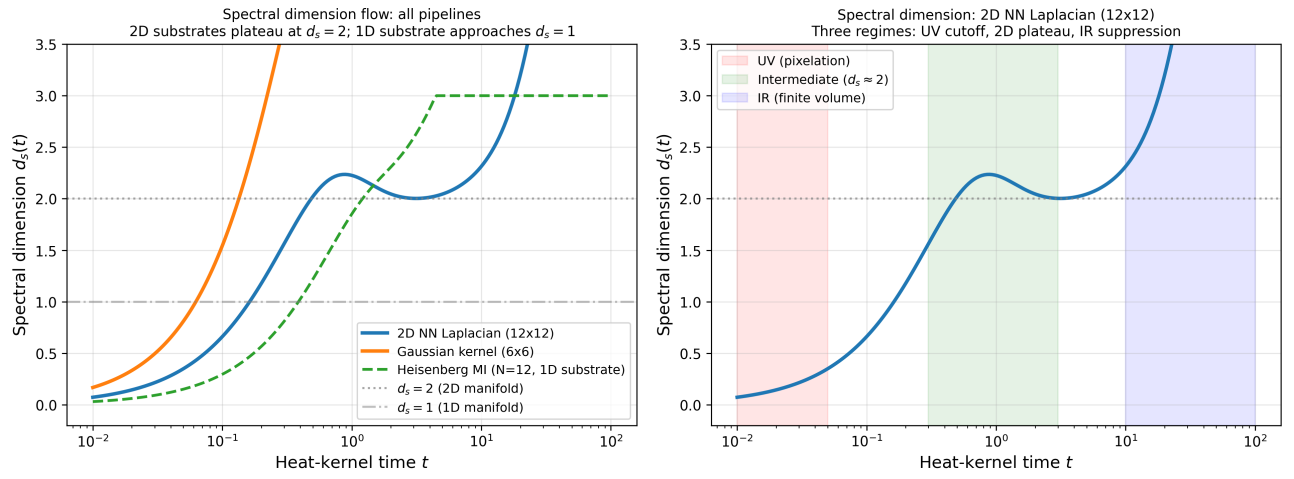


FIG. 5. Spectral dimension flow extracted from the heat-kernel trace. Two-dimensional substrates exhibit a plateau at $d_s \simeq 2$, while the Heisenberg-chain MI Laplacian approaches $d_s \simeq 1$, correctly identifying the underlying dimensionality.

1D Finite-Size Scaling: Continuum Laplace-Beltrami Limit

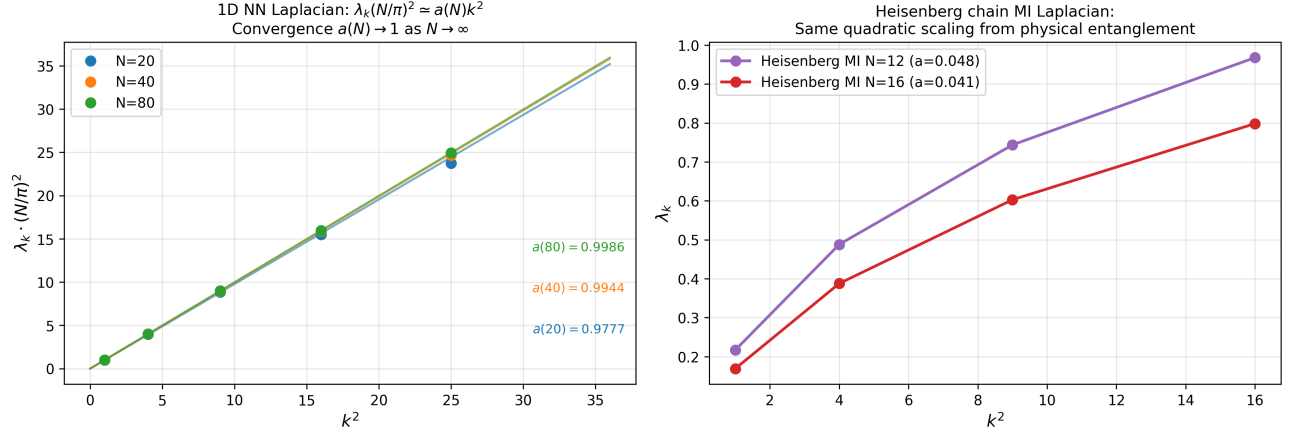


FIG. 6. Mutual-information Laplacian extracted from the antiferromagnetic Heisenberg chain. The quadratic scaling of eigenvalues with mode number matches the continuum Laplace–Beltrami operator on a 1D manifold, demonstrating that physical entanglement alone generates smooth geometry.

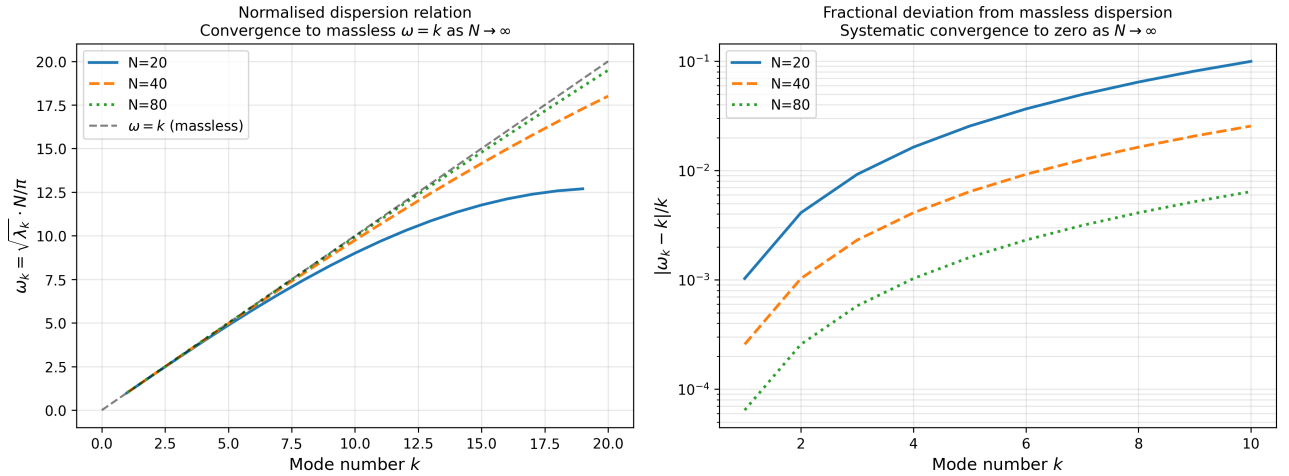


FIG. 7. Normalised dispersion relation $\omega_k = \sqrt{\lambda_k} N/\pi$. The convergence to the massless form $\omega = k$ with $O(1/N^2)$ corrections indicates the presence of graviton-like collective excitations in the continuum limit.